

2520090088(B.DHARSHAN)

SKILL 14

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root@PC: ~/os_lab/append_stderr_lab
ls: cannot access 'non_existent_folder_xyz': No such file or directory
cat: invalid_file_abc.txt: No such file or directory
-----

--- Test 2.3: Capture Execution Failure ---
--- Captured Error Stream (error_capture.log) ---
ls: cannot access 'non_existent_folder_xyz': No such file or directory
cat: invalid_file_abc.txt: No such file or directory
execvp error: No such file or directory
-----

root@PC:~/os_lab/append_stderr_lab# valgrind --leak-check=full --track-fds=yes ./append_redir
valgrind --leak-check=full --track-fds=yes ./stderr_redir
==544== Memcheck, a memory error detector
==544== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==544== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==544== Command: ./append_redir
==544==
=== Task 1: Append Operator Test (>>) ===

--- Running Pass 1: Initial Entry ---
=== Log File Contents (append_log.txt) ===
Entry 1: System Booted At Timestamp 1000
=====

--- Running Pass 2: Appending Second Entry ---
=== Log File Contents (append_log.txt) ===
Entry 1: System Booted At Timestamp 1000
Entry 2: User 'lab_user' Authenticated
=====

--- Running Pass 3: Appending Third Entry ---
=== Log File Contents (append_log.txt) ===
Entry 1: System Booted At Timestamp 1000
Entry 2: User 'lab_user' Authenticated
Entry 3: Shutdown Sequence Initialized
=====
==544==
==544== FILE DESCRIPTORS: 3 open (3 std) at exit.
==544==
```

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root@PC: ~/os_lab/append_stderr_lab
root@PC:~# cd ~/os_lab
mkdir -p append_stderr_lab && cd append_stderr_lab
root@PC:~/os_lab/append_stderr_lab# nano append_redir.c
root@PC:~/os_lab/append_stderr_lab# gcc -g -O0 -Wall -pthread append_redir.c -o append_redir
./append_redir
=== Task 1: Append Operator Test (>>) ===

--- Running Pass 1: Initial Entry ---
=== Log File Contents (append_log.txt) ===
Entry 1: System Booted At Timestamp 1000
=====

--- Running Pass 2: Appending Second Entry ---
=== Log File Contents (append_log.txt) ===
Entry 1: System Booted At Timestamp 1000
Entry 2: User 'lab_user' Authenticated
=====

--- Running Pass 3: Appending Third Entry ---
=== Log File Contents (append_log.txt) ===
Entry 1: System Booted At Timestamp 1000
Entry 2: User 'lab_user' Authenticated
Entry 3: Shutdown Sequence Initialized
=====
root@PC:~/os_lab/append_stderr_lab# nano stderr_redir.c
root@PC:~/os_lab/append_stderr_lab# gcc -g -O0 -Wall -pthread stderr_redir.c -o stderr_redir
./stderr_redir
=== Task 2: Stderr Redirection Test (2> and 2>>) ===

--- Test 2.1: Overwrite Stderr (ls missing_file 2> error_capture.log) ---
--- Captured Error Stream (error_capture.log) ---
ls: cannot access 'non_existent_folder_xyz': No such file or directory
-----

--- Test 2.2: Append Stderr (cat missing_file 2>> error_capture.log) ---
--- Captured Error Stream (error_capture.log) ---
ls: cannot access 'non_existent_folder_xyz': No such file or directory
cat: invalid_file_abc.txt: No such file or directory

```

```
root@PC: ~/os_lab/append_stderr_lab

==544== HEAP SUMMARY:
==544==    in use at exit: 0 bytes in 0 blocks
==544==   total heap usage: 7 allocs, 7 frees, 14,728 bytes allocated
==544==
==544== All heap blocks were freed -- no leaks are possible
==544==
==544== For lists of detected and suppressed errors, rerun with: -s
==544== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
==548== Memcheck, a memory error detector
==548== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==548== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==548== Command: ./stderr_redir
==548==
=== Task 2: Stderr Redirection Test (2> and 2>>) ===

--- Test 2.1: Overwrite Stderr (ls missing_file 2> error_capture.log) ---
--- Captured Error Stream (error_capture.log) ---
ls: cannot access 'non_existent_folder_xyz': No such file or directory
-----

--- Test 2.2: Append Stderr (cat missing_file 2>> error_capture.log) ---
--- Captured Error Stream (error_capture.log) ---
ls: cannot access 'non_existent_folder_xyz': No such file or directory
cat: invalid_file_abc.txt: No such file or directory
-----

--- Test 2.3: Capture Execution Failure ---
==551==
==551== FILE DESCRIPTORS: 4 open (3 std) at exit.
==551== Open file descriptor 3: /dev/pts/0
==551==   at 0x4976C0B: dup (syscall-template.S:120)
==551==   by 0x1094E2: execute_stderr_redirection (stderr_redir.c:36)
==551==   by 0x1098D1: main (stderr_redir.c:130)
==551==
==551==
==551== HEAP SUMMARY:
==551==    in use at exit: 0 bytes in 0 blocks
==551==   total heap usage: 5 allocs, 5 frees, 10,160 bytes allocated
==551==
==551== All heap blocks were freed -- no leaks are possible
```

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root@PC: ~/os_lab/append_stderr_lab

==551==
==551== HEAP SUMMARY:
==551==    in use at exit: 0 bytes in 0 blocks
==551==   total heap usage: 5 allocs, 5 frees, 10,160 bytes allocated
==551==
==551== All heap blocks were freed -- no leaks are possible
==551==
==551== For lists of detected and suppressed errors, rerun with: -s
==551== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
--- Captured Error Stream (error_capture.log) ---
ls: cannot access 'non_existent_folder_xyz': No such file or directory
cat: invalid_file_abc.txt: No such file or directory
execvp error: No such file or directory
-----
==548==
==548== FILE DESCRIPTORS: 3 open (3 std) at exit.
==548==
==548== HEAP SUMMARY:
==548==    in use at exit: 0 bytes in 0 blocks
==548==   total heap usage: 7 allocs, 7 frees, 14,728 bytes allocated
==548==
==548== All heap blocks were freed -- no leaks are possible
==548==
==548== For lists of detected and suppressed errors, rerun with: -s
==548== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
root@PC:~/os_lab/append_stderr_lab# strace -f -e trace=openat,dup2,close ./stderr_re
dir
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
close(3)                                = 0
=== Task 2: Stderr Redirection Test (2> and 2>>) ===

--- Test 2.1: Overwrite Stderr (ls missing_file 2> error_capture.log) ---
openat(AT_FDCWD, "error_capture.log", O_WRONLY|O_CREAT|O_TRUNC, 0644) = 4
dup2(4, 2)                        = 2
close(4)                          = 0
strace: Process 556 attached
[pid  556] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 4
[pid  556] close(4)                        = 0
```

```
root@PC: ~/os_lab/append_stderr_lab
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NAME", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NAME", O_RDONLY|O_CLOEXEC) =
4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_PAPER", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_PAPER", O_RDONLY|O_CLOEXEC)
= 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES", O_RDONLY|O_CLOEX
EC) = -1 ENOENT (No such file or directory)
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES", O_RDONLY|O_CLOEXE
C) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES", O
_RDONLY|O_CLOEXEC) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MONETARY", O_RDONLY|O_CLOEX
EC) = -1 ENOENT (No such file or directory)
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MONETARY", O_RDONLY|O_CLOEXE
C) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_COLLATE", O_RDONLY|O_CLOEXE
C) = -1 ENOENT (No such file or directory)
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_COLLATE", O_RDONLY|O_CLOEXEC
) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TIME", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TIME", O_RDONLY|O_CLOEXEC) =
4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NUMERIC", O_RDONLY|O_CLOEXE
C) = -1 ENOENT (No such file or directory)
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NUMERIC", O_RDONLY|O_CLOEXEC
) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_CTYPE", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)
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root@PC: ~/os_lab/append_stderr_lab
) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_CTYPE", O_RDONLY|O_CLOEXEC)
= -1 ENOENT (No such file or directory)
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_CTYPE", O_RDONLY|O_CLOEXEC)
= 4
[pid 556] close(4) = 0
[pid 556] close(1) = 0
[pid 556] close(2) = 0
[pid 556] +++ exited with 2 +++
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=556, si_uid=0, si_status=2
, si_utime=0, si_stime=1 /* 0.01 s */} ---
dup2(3, 2) = 2
close(3) = 0
--- Captured Error Stream (error_capture.log) ---
openat(AT_FDCWD, "error_capture.log", O_RDONLY) = 3
ls: cannot access 'non_existent_folder_xyz': No such file or directory
close(3) = 0
-----

--- Test 2.2: Append Stderr (cat missing_file 2>> error_capture.log) ---
openat(AT_FDCWD, "error_capture.log", O_WRONLY|O_CREAT|O_APPEND, 0644) = 4
dup2(4, 2) = 2
close(4) = 0
strace: Process 557 attached
[pid 557] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 4
[pid 557] close(4) = 0
[pid 557] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC)
= 4
[pid 557] close(4) = 0
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/locale-archive", O_RDONLY|O_CLOEXEC) =
-1 ENOENT (No such file or directory)
[pid 557] openat(AT_FDCWD, "/usr/share/locale/locale.alias", O_RDONLY|O_CLOEXEC) =
4
[pid 557] close(4) = 0
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_IDENTIFICATION", O_RDONLY|O
_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_IDENTIFICATION", O_RDONLY|O
_CLOEXEC) = 4
[pid 557] close(4) = 0

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root@PC: ~/os_lab/append_stderr_lab
strace: Process 556 attached
[pid 556] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/locale-archive", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 556] openat(AT_FDCWD, "/usr/share/locale/locale.alias", O_RDONLY|O_CLOEXEC) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache", O_RDONLY|O_CLOEXEC) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = 4
[pid 556] close(4) = 0
[pid 556] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_ADDRESS", O_RDONLY|O_CLOEXEC)
```

root@PC: ~/os_lab/append_stderr_lab

```
O_RDONLY|O_CLOEXEC) = 4
[pid 557] close(4) = 0
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = 4
[pid 557] close(4) = 0
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = 4
[pid 557] close(4) = 0
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = 4
[pid 557] close(4) = 0
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NAME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NAME", O_RDONLY|O_CLOEXEC) = 4
[pid 557] close(4) = 0
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_PAPER", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_PAPER", O_RDONLY|O_CLOEXEC) = 4
[pid 557] close(4) = 0
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 4
[pid 557] close(4) = 0
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 4
[pid 557] close(4) = 0
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 557] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = 4
[pid 557] close(4) = 0
```

```
root@PC: ~/os_lab/append_stderr_lab
= 4
[pid 557] close(4) = 0
[pid 557] openat(AT_FDCWD, "invalid_file_abc.txt", O_RDONLY) = -1 ENOENT (No such
file or directory)
[pid 557] close(1) = 0
[pid 557] close(2) = 0
[pid 557] +++ exited with 1 +++
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=557, si_uid=0, si_status=1
, si_ftime=0, si_stime=1 /* 0.01 s */} ---
dup2(3, 2) = 2
close(3) = 0
--- Captured Error Stream (error_capture.log) ---
openat(AT_FDCWD, "error_capture.log", O_RDONLY) = 3
ls: cannot access 'non_existent_folder_xyz': No such file or directory
cat: invalid_file_abc.txt: No such file or directory
close(3) = 0
-----

--- Test 2.3: Capture Execution Failure ---
openat(AT_FDCWD, "error_capture.log", O_WRONLY|O_CREAT|O_APPEND, 0644) = 4
dup2(4, 2) = 2
close(4) = 0
strace: Process 558 attached
[pid 558] close(4) = 0
[pid 558] +++ exited with 1 +++
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=558, si_uid=0, si_status=1
, si_ftime=0, si_stime=0} ---
dup2(3, 2) = 2
close(3) = 0
--- Captured Error Stream (error_capture.log) ---
openat(AT_FDCWD, "error_capture.log", O_RDONLY) = 3
ls: cannot access 'non_existent_folder_xyz': No such file or directory
cat: invalid_file_abc.txt: No such file or directory
execvp error: No such file or directory
close(3) = 0
-----
+++ exited with 0 +++
root@PC:~/os_lab/append_stderr_lab#
```